AE462 Lab 6: Supersonic Flow over Diamond Airfoil

Lab Session: Wednesday, March 25, 2026
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Introduction

This laboratory assignment will focus on setting up solver configurations, mesh generation via Python, and simulating supersonic flows. This laboratory exercise considers a diamond airfoil, having thickness of $t/c = 0.04$, in a $M = 1.7$ freestream flow. This airfoil configuration produces a surface half-angle oriented 2.29 relative to the chord line.

The coordinates of the airfoil are given by the following relationship (+ for the upper surface, − for the lower surface):

$$\frac{y}{c} = \pm \begin{cases} 0.04\,\dfrac{x}{c}, & 0 \le \dfrac{x}{c} < 0.5 \\ 0.04\left(1 - \dfrac{x}{c}\right), & 0.5 \le \dfrac{x}{c} \le 1 \end{cases}$$

The mesh generated in Python included 200 grid points per edge of the diamond airfoil and 400 nodes in the far field, with a far-field radius of 80 m.

I. Comparison with Theory

Shock-expansion theory is often used to analyze airfoils in supersonic flow. An approximate linear theory can also be developed assuming small angles and sufficiently weak Mach waves (which are both valid assumptions in this exercise).

A. Pressure Coefficient Distribution

For the baseline case of $M_\infty = 1.7$ and $\alpha = 1°$, the pressure-coefficient distribution predicted by linear theory was compared against the Euler CFD solution. Figure 1 shows the resulting co-plot of the upper and lower surface C_p distributions.

B. Comparison of Aerodynamic Coefficients

The lift and drag coefficients from linear theory and Euler CFD were compared for the same baseline case. These values are summarized in Table 1.

Table 1 Comparison of aerodynamic coefficients from linear theory and Euler CFD at $M_\infty = 1.7$ and $\alpha = 1°$.

Method	C_l	C_d
Linear Theory	0.050782	0.005542
Euler CFD	0.050655	0.005488

The C_l and C_d values are very similar to each other, with the Euler values slightly less than the Linear Theory ones.

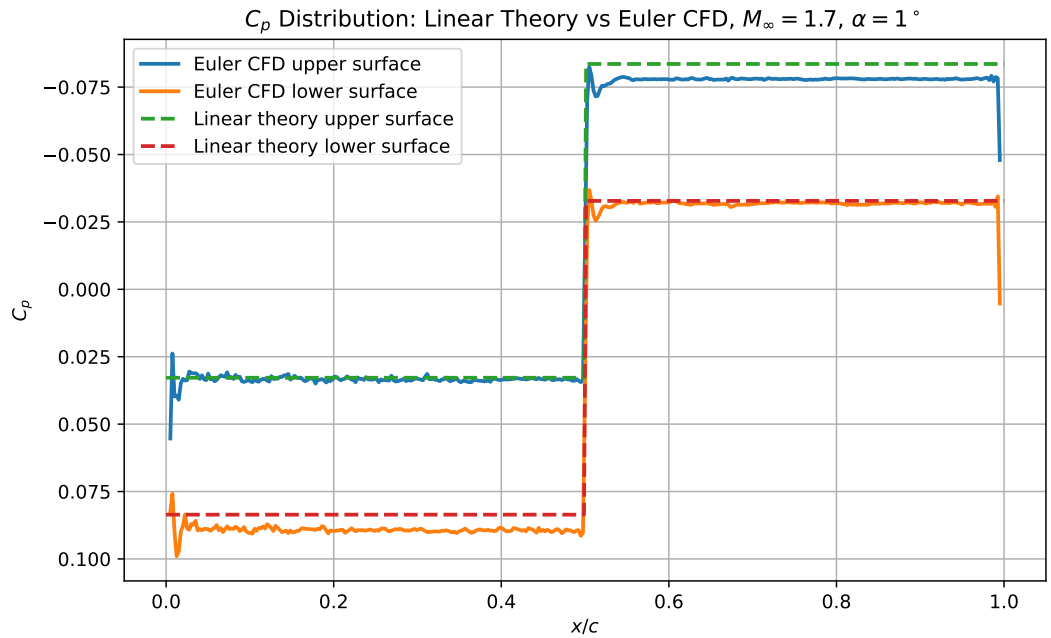

Fig. 1 C_p distribution comparison between linear theory and Euler CFD for the diamond airfoil at $M_\infty = 1.7$ and $\alpha = 1^\circ$.

II. Aerodynamic Polar

A. Euler CFD Polar

The aerodynamic polar of the diamond airfoil was generated using Euler CFD solutions across an angle-of-attack range from $\alpha = 0^\circ$ to 10° in increments of 2°. The lift, drag, and pitching moment coefficients are shown in Figures 2, 3, and 4, respectively.

B. Lift-Curve Slope

The lift-curve slope was determined from the Euler CFD lift polar over the linear angle-of-attack range. This value was then compared with the prediction from linear theory,

$$C_{l,\alpha} = \frac{4}{\sqrt{M_\infty^2 - 1}}.$$

The resulting values are summarized in Table 2.

Table 2 Comparison of lift-curve slope from Euler CFD and linear theory.

Method	$C_{l,\alpha}$ (1/rad)
Linear Theory	2.909572
Euler CFD	2.972427

The Euler CFD lift slope is 2.16% greater than the linear theory one. These two values are very comparable, and depending on which points set as within the linear region, was closer to 0.11% off. This value was choosing $\alpha = 6$ or lower as the linear region. However, since the slope doesn't fall off, all the alphas are still within the linear region.

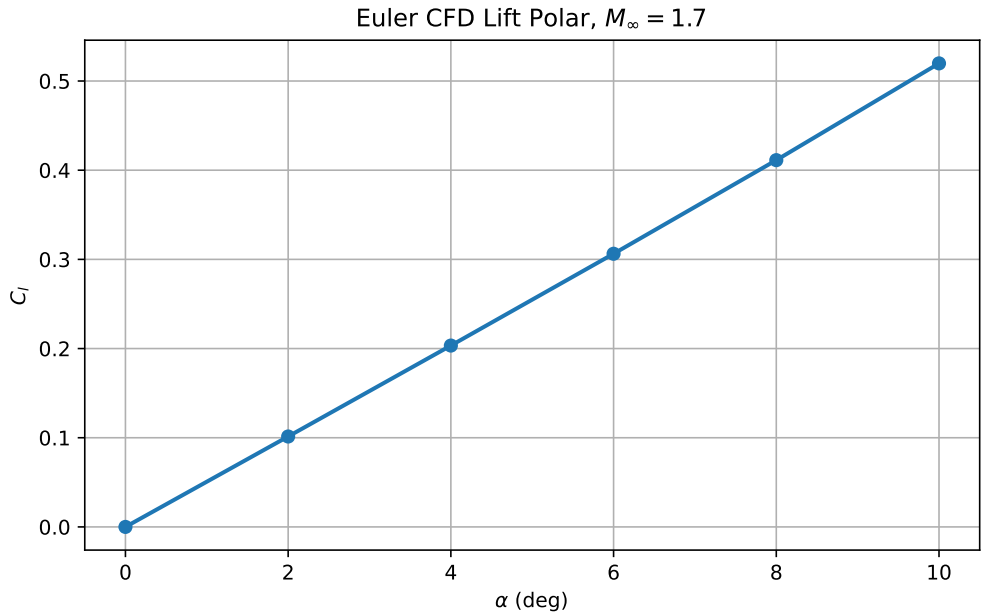

Fig. 2 Lift coefficient variation with angle of attack from Euler CFD.

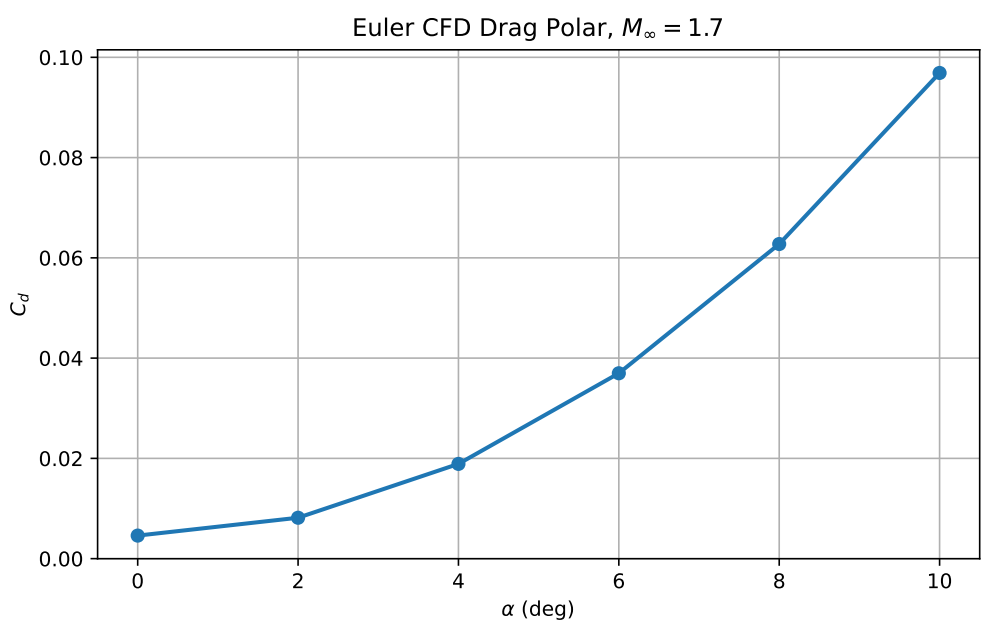

Fig. 3 Drag coefficient variation with angle of attack from Euler CFD.

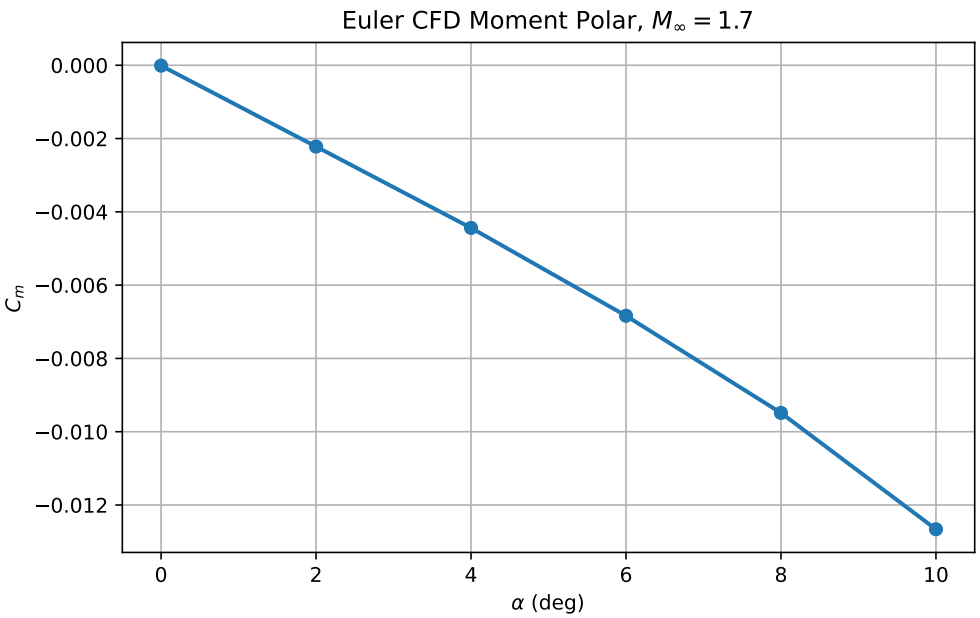

Fig. 4 Moment coefficient variation with angle of attack from Euler CFD.

III. Viscous Flow Analysis

A. Pressure Coefficient Comparison

A viscous RANS solution was computed for the same baseline case, $M_\infty = 1.7$ and $\alpha = 1°$, and the resulting C_p distribution was compared against the inviscid Euler solution. Figure 5 shows the co-plot of the upper and lower surface pressure distributions for both methods.

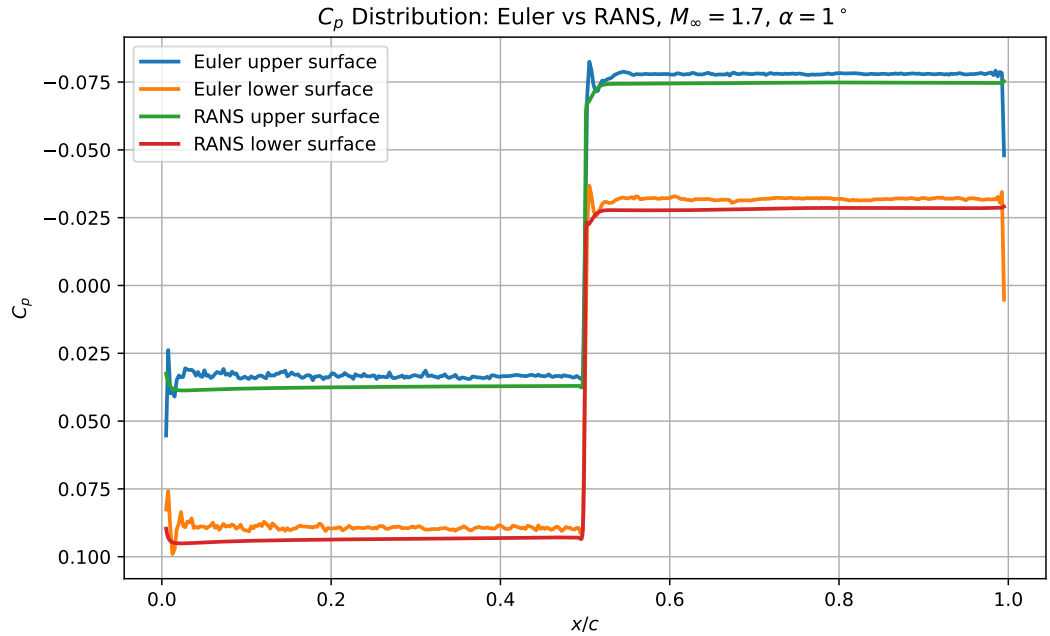

Fig. 5 C_p distribution comparison between Euler and RANS solutions for the diamond airfoil at $M_\infty = 1.7$ and $\alpha = 1°$.

B. Comparison of Aerodynamic Coefficients

The lift and drag coefficients from the viscous RANS and inviscid Euler simulations are summarized in Table 3.

Table 3 Comparison of aerodynamic coefficients from Euler and RANS CFD at $M_\infty = 1.7$ and $\alpha = 1°$.

Method	C_l	C_d
Euler CFD	0.050655	0.005488
RANS CFD	0.050975	0.009624

The lift coefficients are relatively close, while the drag coefficient from the RANS solution is 75.35% greater.

C. Discussion of Drag Difference

The RANS solution predicts a larger drag coefficient than the Euler solution because viscous effects are included in the governing equations. In the Euler formulation, viscosity is neglected, so the solution does not capture skin-friction drag or boundary-layer losses. The RANS simulation includes viscous shear stresses and therefore predicts a higher total drag.

IV. Conclusion

This lab compared linearized supersonic theory, inviscid Euler CFD, and viscous RANS CFD for a diamond airfoil in supersonic flow. The Euler results showed generally good agreement with linear theory for the baseline case, while

the aerodynamic polar demonstrated the expected trends in lift, drag, and moment with increasing angle of attack. The viscous RANS solution further showed that including viscosity has only a modest effect on the lift coefficient at this condition, but produces a noticeable increase in drag due to boundary-layer and skin-friction effects.

Appendix

The GitHub repository containing all simulation files and post-processing scripts is available at:
`https://github.com/nikitaaa27/AE462-git/tree/main/Lab6`